17} Caesarian Section

Goals:

to improve patient safety and outcomes, enhance the training of healthcare professionals, and optimize the efficiency of the procedure, developing a comprehensive understanding of the surgical steps, implementing evidence-based practices, improving interprofessional collaboration, reducing maternal and neonatal complications, standardizing techniques for consistency and training, and evaluating the effectiveness of new training methods and guidelines

Objectives:

By the end of this procedure, the student nurse should be able to:

- ☞ Recognize common indications for cesarean delivery
- ☞ Select optimal clinical approaches based on current evidence and patient-specific factors.
- ☞ Apply evidence-based practices in all phases of the delivery process to minimize complications like infection and improve patient recovery.
- ☞ Promote interprofessional collaboration:
- ☞ Develop and adopt standardized, evidence-based methods for clinical steps.

Outlines:

- ☞ Introduction
- ☞ Definition.
- ☞ indication
- ☞ Equipment.
- ☞ Procedure.

Introduction:

A cesarean section involves making incisions through the mother's abdomen and uterus to deliver the baby. The steps include anesthesia, surgical preparation, abdominal and uterine incisions, delivery of the baby and placenta, administering an oxytocin injection, and then closing all the layers of the abdomen and uterus with stitches or staples

Definition:

Cesarean Section is the surgical delivery of the fetus by incisions in the pennant's abdomen and in the uterus

Indications:

1. Placenta previa.
2. Cephalopelvic disproportion (CPD): CPD is a term that means fetus's s head or body is too large to pass safely through pelvis, or pelvis is too small to deliver an average-sized baby.
3. Previous C-section: Although it's possible to have a vaginal birth after a previous C-section, it's not an option for everyone. Factors that can affect this include the type of uterine incision used in the previous C-section and the risk of uterine rupture.
4. Multiples pregnancies: Although twins can often be 70% delivered vaginally, two or more babies might require a C-section.
5. Malpresenation e.g Transverse lie and breech presentation.
6. Health conditions: Conditions like heart disease, genital herpes.

7. Obstruction: A large uterine fibroid, a pelvic fracture or fetus with congenital anomalies.

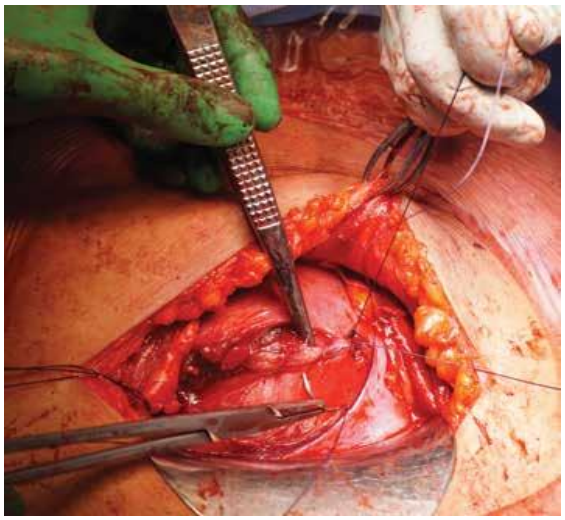
Equipment:

1. Scalpels,
2. Forceps (to hold tissue and stop bleeding),
3. Retractors (to hold the incision open),
4. Scissors (to cut tissue), clamps (to manage tissue and vessels)
5. Suction apparatus,
6. Sterilizers,
7. Drapes,
8. Sutures.
9. Operating table,
10. Iv lines for fluids and medication, and a urinary catheter.

Cesarean section

Steps	RAT IONAL
Getting Ready:	
1. Admit the mother one or two days before the proposed date. 2. Explain the operation 3. Take a complete history 4. Starve (fasting) the mother for 6 hrs. before the operation 5. Perform the investigation. 6. Carry out a full general examination. 7. Sign in the consent form. 8. Remove all jewelers or denatures and nail polish. 9. Insert a catheter and cannula.	
Management of Cesarean Section Steps	
1. Handle the needed equipment as requested.	
2. IV antibiotics: 2 g IV Cephazolin as routine prophylaxis 15–60 minutes prior to skin incision.	
3. Administer anesthesia, often an epidural or spinal block and medication as prescribed.	
4. Observe carefully the conscious level.	
5. Prepare the skin with an antiseptic solution e.g. chlorhexidine-alcohol, which should be allowed to dry prior to draping.	It is important that solution does dry and doesn't pool underneath the drapes, as this is a fire risk and patient burns have occurred previously.

6. **Skin incision**: ☞ Make a straight incision 3 cm below the level of the anterior superior iliac spines. ☞ Entry technique: sharp entry through the skin, middle 3 cm of the subcutaneous fat and rectus sheath is demonstrated here. ☞ Blunt extension of the subcutaneous tissue and rectus sheath. ☞ Blunt entry into the peritoneal cavity high in order to avoid entering into the bladder.	
7. **Uterine Incision:** Make a small 2–3 cm horizontal sharp incision in the lower uterine segment, (usually a low transverse horizontal) incision.	
8. Baby Delivery:	

✍ Deliver the baby through the incisions, often within 5 to 15 minutes. ☞ Delivery of the fetal head with flexion and elevation of the fetal head toward the uterine incision. ☞ Make fundal pressure to deliver rest of fetus body. ✍ Apply immediate baby care. ☞ Clear the baby's mouth and nose. ☞ Clamp the umbilical cord and cut. **4. Placenta Removal:** ☞ Fundal massage and controlled cord traction. ☞ Wait until spontaneous delivery of the placenta. **5. Oxytocin Injection:** Give oxytocin injection to help the uterus contract and reduce bleeding. **6. Closer** ☞ Identifying the uterine incision and uterine angles. ☞ Close the layers of muscle, fat, and skin of the abdomen, as well as the uterus, with a double-layer, non-locking continuous monofilament (1 monocryl) sutures or staples.	- An oxytocic is usually given following the delivery of the fetus to reduce the risk of PPH. 

- Routinely checking the tubes and ovaries.
- Check haemostasis between the rectus sheath and muscle for injury to perforating vessels during entry.

Dressing:

A dressing is applied over the incision.

Recovery:

- Hold newborn soon after delivery, and encourage skin-to-skin contact.
- Pain management, deep breathing exercises, and wound care.

Nursing Responsibilities in the operating room (O.R):

1. Informing the physician of complications and calling the anesthesiologist.
2. Notifying the pediatrician and resuscitation team.
3. Informed consent must be obtained.
4. Insert Foley catheter.
5. Administering ordered medications.
6. Take patient to the OR and transfer to the operating table.

7. Positioning the patient. 8. Assuring Infant resuscitation equipment 9. Obtaining a fetal heart rate. 10. Placing a grounding pad under her thigh and attaching the electro cautery unit. 11. Preparing the patient's abdomen for surgery. 12. Counting instruments; sponges and sharps prior to surgery. 13. The circulating nurse is responsible for making sure the counts of these items are accurate.	
Post Procedure Tasks:	

1. Assess vital signs & consciousness. 2. Observe and assess the wound or dressing site for bleeding. 3. Observe the lochia for any excessive bleeding. 4. Bring the baby to the breast as soon as possible after the mother wakes up. 5. When the bowel sounds are present and the mother condition is satisfactory, discontinue IV fluids and commence oral fluids. 6. Remove the urinary catheter after 6 hrs postoperative, sometimes depends on mother's condition. 7. Assist the mother during breast feeding. 8. Give light diet and then a normal diet as soon as she can tolerate it. 9. Encourage her to walk when possible after measuring her blood pressure. 10. Discharge the mother from hospital on the 3rd or 5th day. 11. Instruct her to come to remove the sutures as doctor order (on 7th or 8th day). 12. <u>Give health teaching and discharge Instructions for the Patient</u> as: - No driving for 10 days.	

✍ Refrain from intercourse for 4 to 6 weeks. ✍ Water can wash over the wound as long as water has no direct impact on the wound. ✍ Keep the wound clean and dry. ✍ Notify the physician's office of the following problems: Pus seeping out of the wound, fever, painful urination, difficulty breathing, shortness of breath, or increasing pain. ✍ Follow up in the office for a wound check in 1 week. ✍ Limit activity for the first week, back to full activity by 6 weeks.	

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	Done	Not done	Done	Not done
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Total score:				
Date:				

Student name: **Instructor name:**